

|   |        |                                                                   |    |     |
|---|--------|-------------------------------------------------------------------|----|-----|
|   |        | amplitude of second about twice amplitude of first .....          | B1 | [3] |
| 7 | (a)(i) | energy required to separate the nucleons in a nucleus .....       | M1 |     |
|   |        | nucleons separated to infinity / completely .....                 | A1 | [2] |
|   | (ii)   | S shown at peak .....                                             | B1 | [1] |
|   | (b)(i) | 4 .....                                                           | A1 | [1] |
|   | (ii)1. | idea of energy as product of $A$ and energy per nucleon .....     | C1 |     |
|   |        | energy = $(8.37 \times 142 + 8.72 \times 90) - 235 \times 7.59$   |    |     |
|   |        | = $1189 + 785 - 178$                                              |    |     |
|   |        | = 190 MeV .....(-1 for each a.e.) .....                           | A2 | [3] |
|   | 2.     | energy = $mc^2$ .....                                             | C1 |     |
|   |        | 1 MeV = $1.6 \times 10^{-13}$ J .....                             | C1 |     |
|   |        | energy = $(190 \times 1.6 \times 10^{-13}) / (3.0 \times 10^8)^2$ |    |     |
|   |        | = $3.4 \times 10^{-28}$ kg .....                                  | A1 | [3] |